COMPASS Synoptic: Porewater SO4/Cl

July 2026 Samples

2026-10-28

Contents

1 Run Information **2**

1.1 Assess Standard Curves . . . 3

1.2 Assess Check Standards . . . 5

1.3 Assess Blanks . . . 6

1.4 Assess Duplicates . . . 7

1.5 Calculate mmol/L concentrations & salinity, add dilutions . . . 9

1.6 Assess Analytical Spikes . . . 9

1.7 Check if samples are within the range of the standard curve . . . 10

1.8 Check to see if samples run match metadata & merge info . . . 10

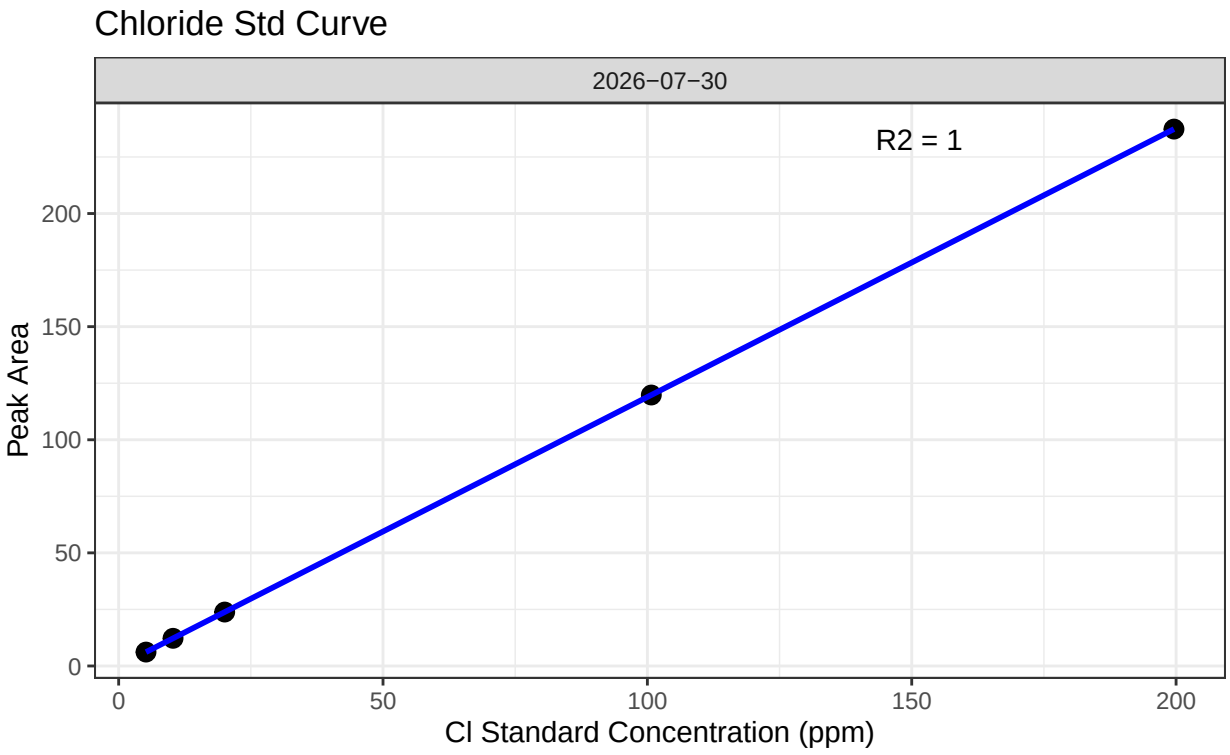
1 Run Information

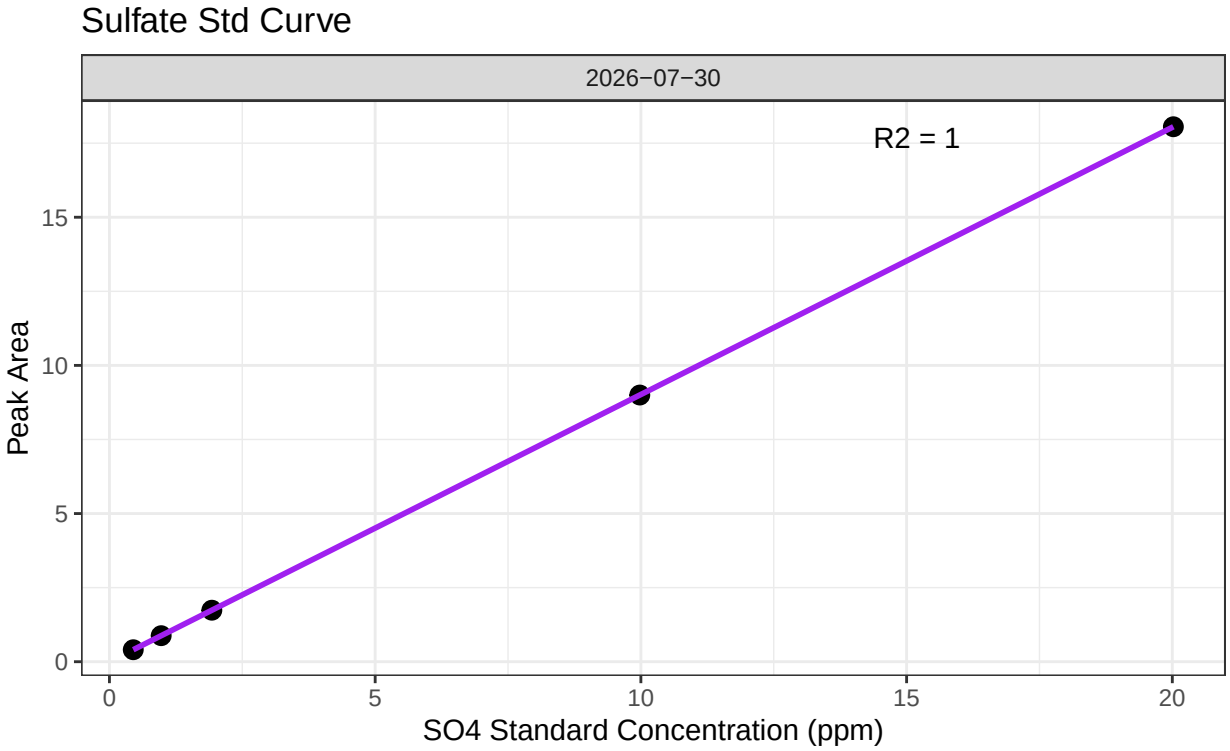
Run date: 2026-07-30
Run by: Zoe Read
Run notes: Run notes

SO4 file used: - COMPASS_Synoptic_CB_MonMon_202607_MSM_GWI_SO4.txt

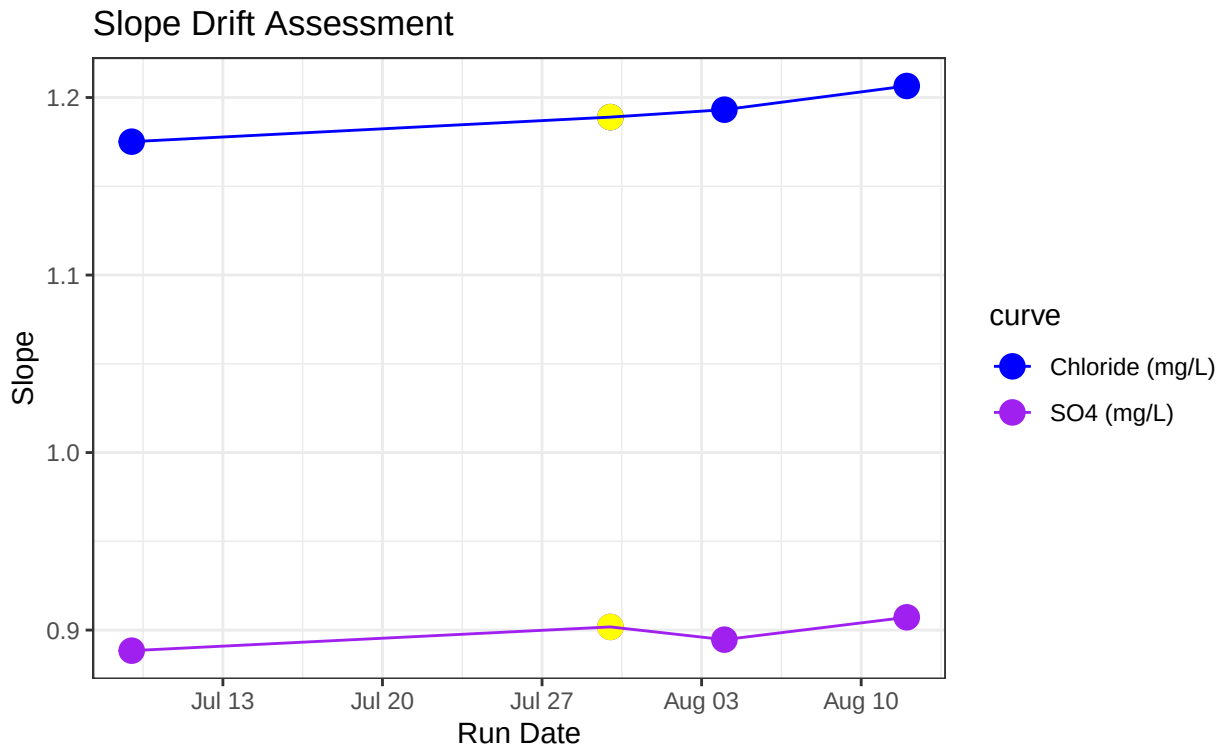
Cl file used: - COMPASS_Synoptic_CB_MonMon_202607_MSM_GWI_Cl.txt

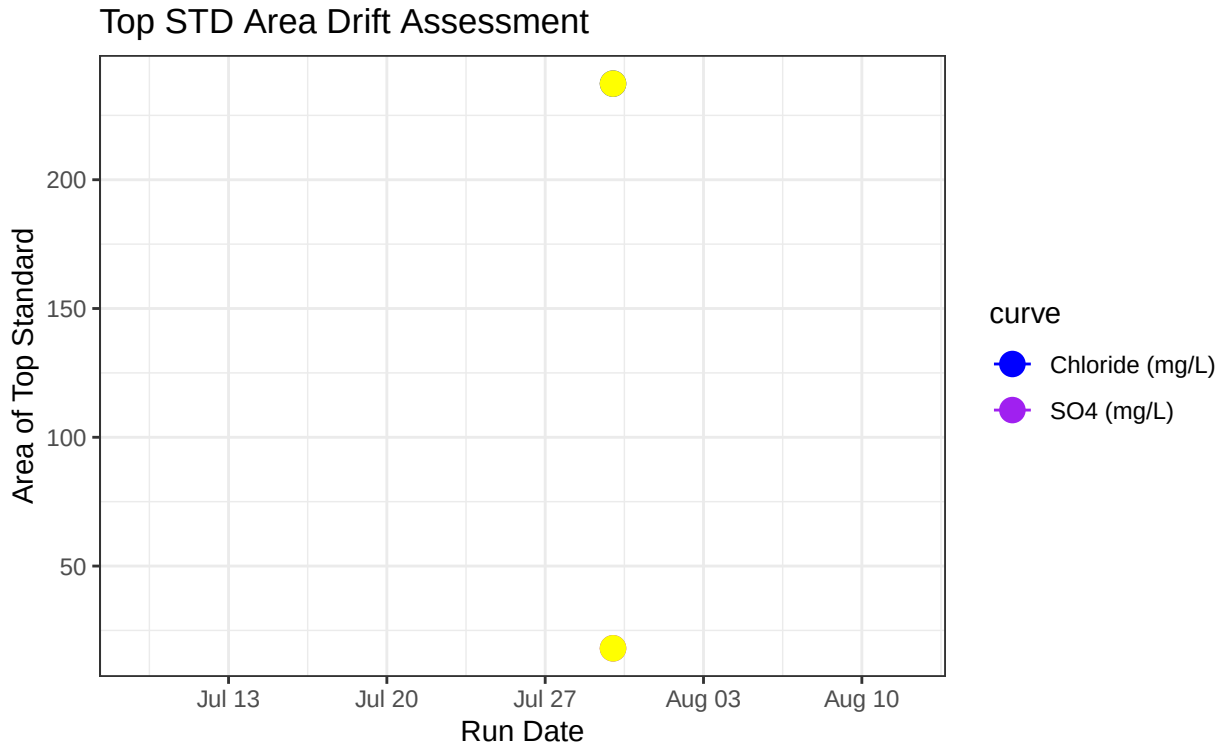
1.1 Assess Standard Curves






```
## [1] "QAQC log file exists and has been read into the code."
```






```
## [1] "Cl Curve r2 GOOD"
```

```
## [1] "SO4 Curve r2 GOOD"
```

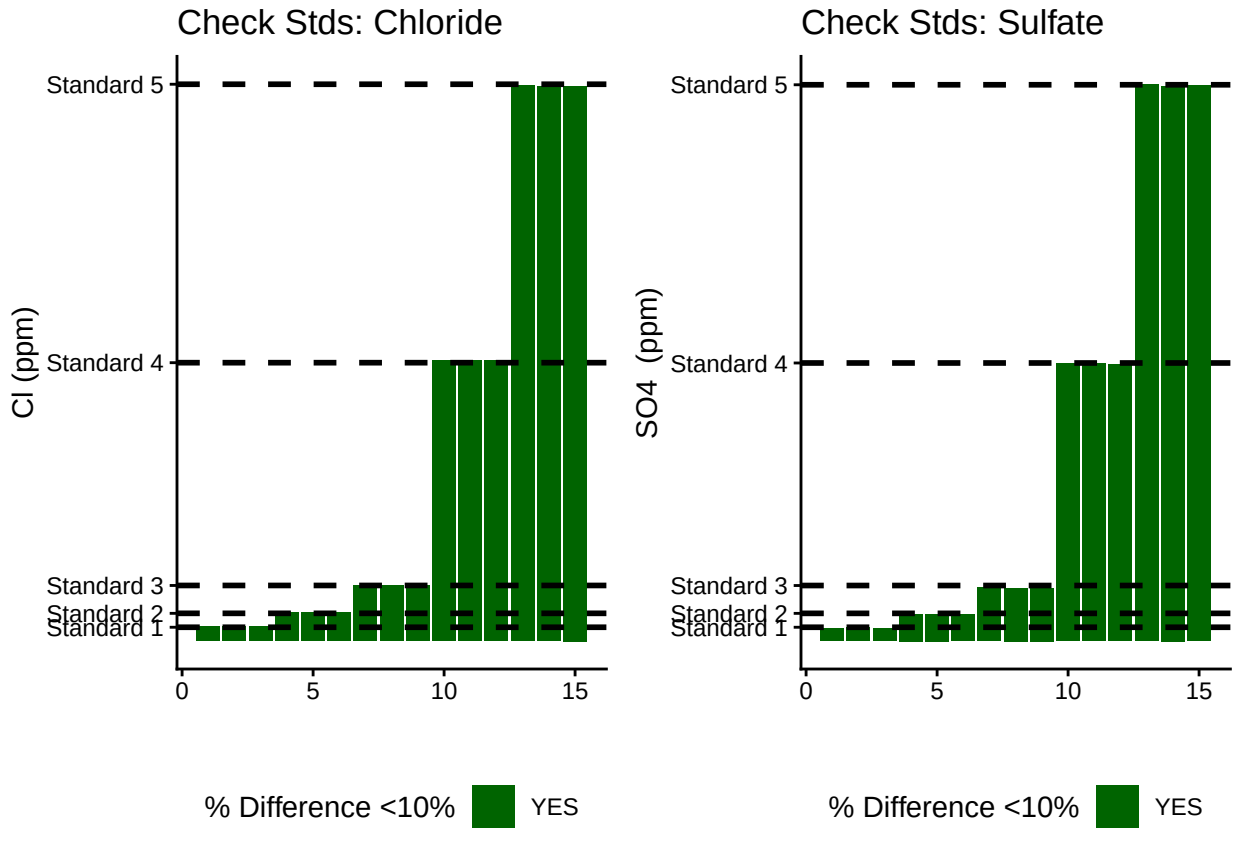
1.2 Assess Check Standards

```
## # A tibble: 5 x 5
##   sample_ID  mean_Cl   sd_Cl    cv_Cl flag_Cl
##   <chr>        <dbl>   <dbl>    <dbl> <chr>
## 1 Standard 1    5.19 0.0218  0.00420  Chloride Check Standard RSD within Range ~
## 2 Standard 2   10.3  0.0192  0.00187  Chloride Check Standard RSD within Range ~
## 3 Standard 3   20.1  0.00968 0.000482 Chloride Check Standard RSD within Range ~
## 4 Standard 4  101.   0.0614  0.000609 Chloride Check Standard RSD within Range ~
## 5 Standard 5  199.   0.194   0.000972 Chloride Check Standard RSD within Range ~
```

```
## # A tibble: 5 x 5
##   sample_ID  mean_SO4  sd_SO4  cv_SO4 flag_SO4
##   <chr>         <dbl>   <dbl>   <dbl> <chr>
## 1 Standard 1    0.459 0.0195  0.0424  Sulfate Check Standard RSD within Range ~~
## 2 Standard 2    0.972 0.00658 0.00677 Sulfate Check Standard RSD within Range ~~
## 3 Standard 3    1.91  0.0114  0.00596 Sulfate Check Standard RSD within Range ~~
## 4 Standard 4    9.97  0.0165  0.00165 Sulfate Check Standard RSD within Range ~~
## 5 Standard 5   20.0   0.0312  0.00156 Sulfate Check Standard RSD within Range ~~
```

```
## [1] ">80% of Chloride Check Standards have RSD within range - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards have RSD within range - PROCEED"
```



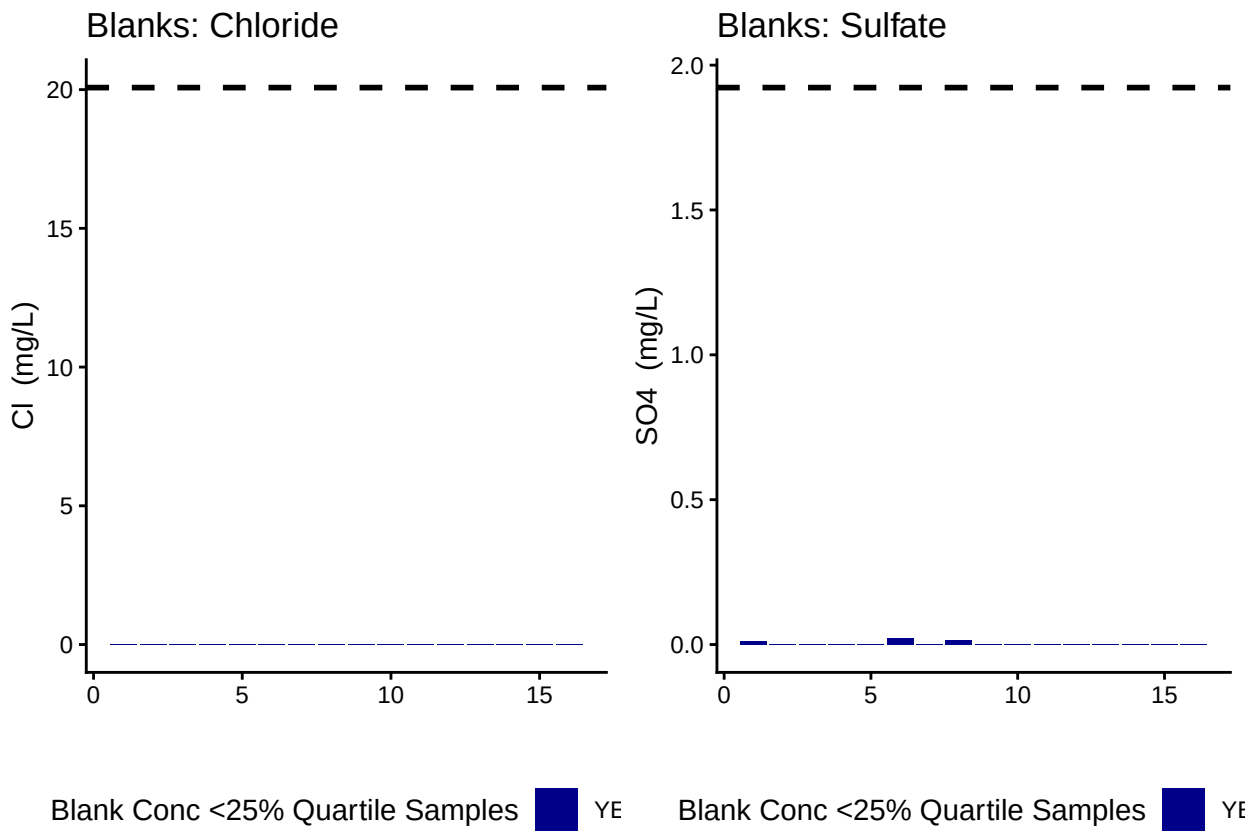

```
## [1] ">80% of Chloride Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">80% of Sulfate Check Standards are within range of expected concentration - PROCEED"
```

1.3 Assess Blanks

```
## [1] ">80% of Chloride Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">80% of Sulfate Blank concentrations are lower 25% quartile of samples"
```



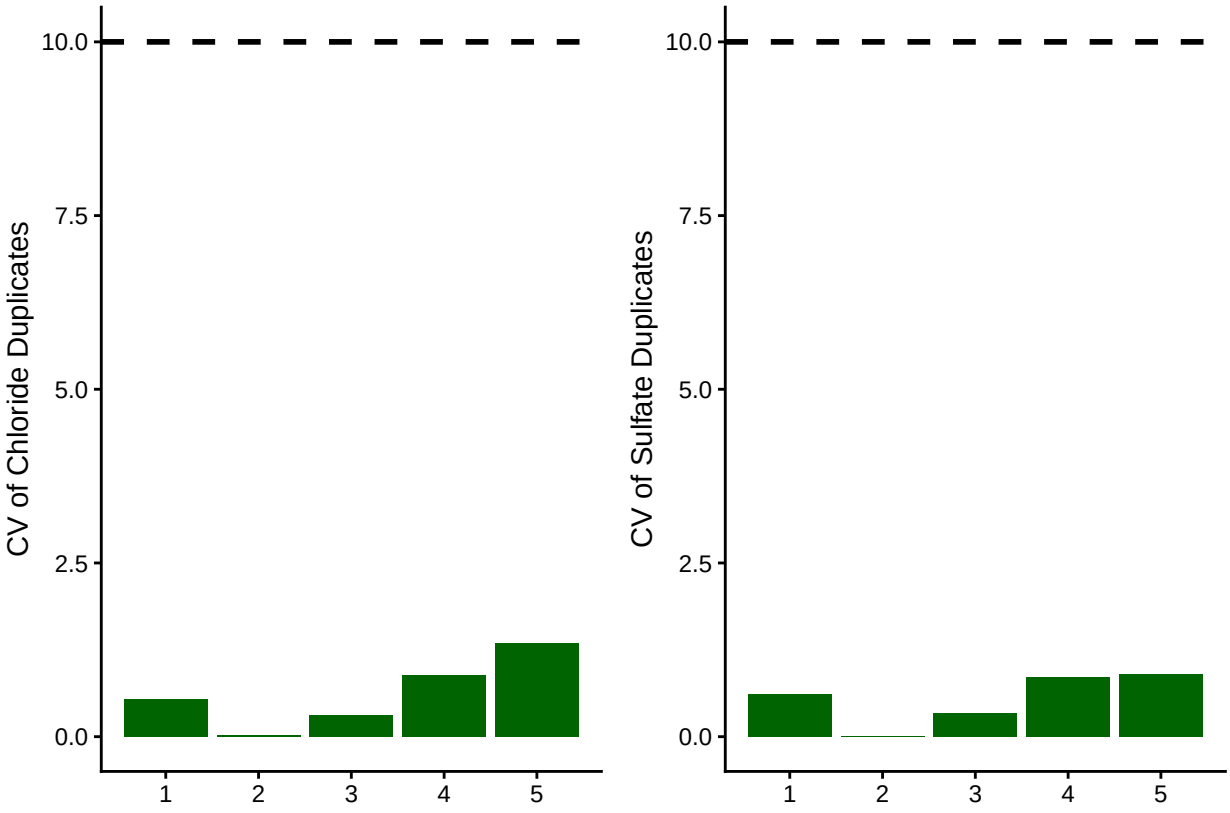

```
## Chloride blanks mean ppm:
```

```
## [1] 0.00820625
```

```
## Sulfate blanks mean ppm:
```

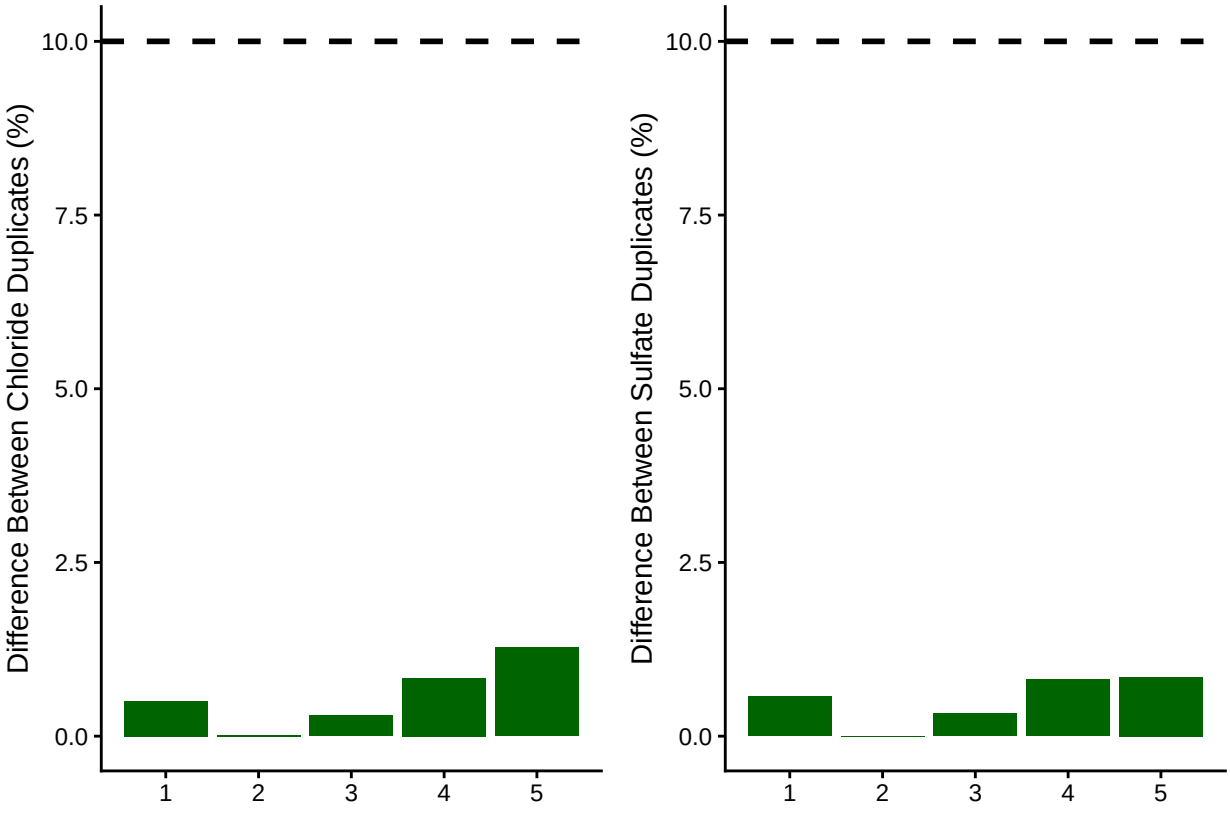
```
## [1] 0.003275
```

1.4 Assess Duplicates




```
## [1] ">80% of Chloride Duplicates have a CV <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a CV <10% - PROCEED"
```

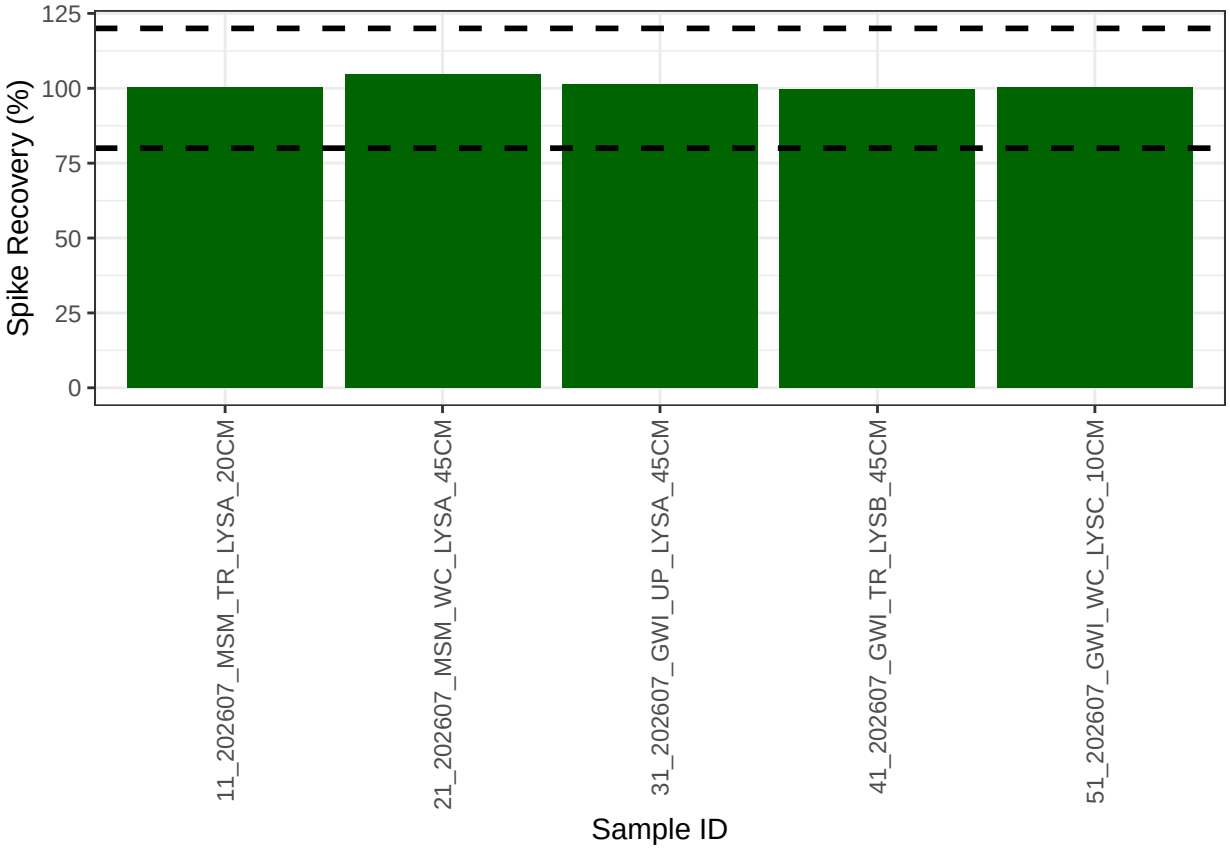



```
## [1] ">80% of Chloride Duplicates have a percent difference <10% - PROCEED"
```

```
## [1] ">80% of Sulfate Duplicates have a percent difference <10% - PROCEED"
```

1.5 Calculate mmol/L concentrations & salinity, add dilutions

1.6 Assess Analytical Spikes




```
## [1] ">80% of SO4 spikes have a recovery between the high and low cutoff - PROCEED"
```

1.7 Check if samples are within the range of the standard curve

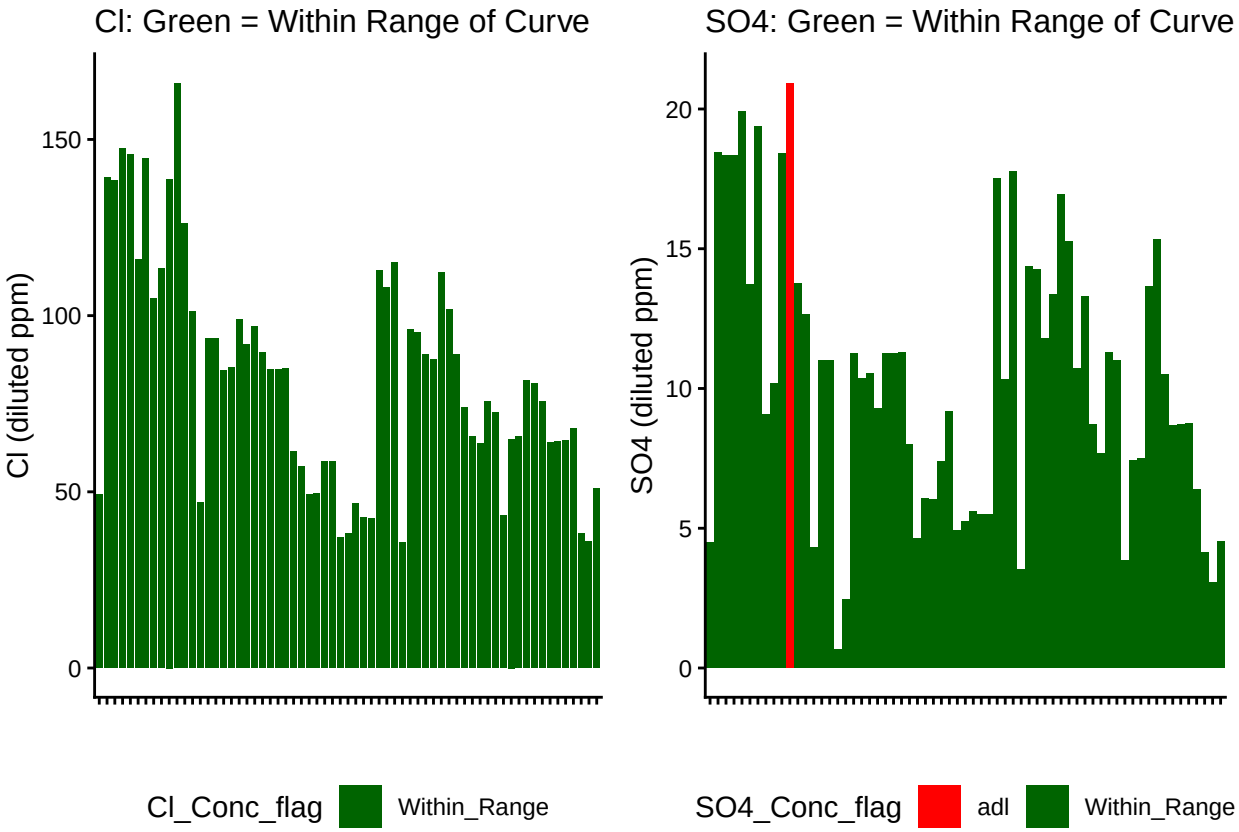


Table 1: SO4 samples

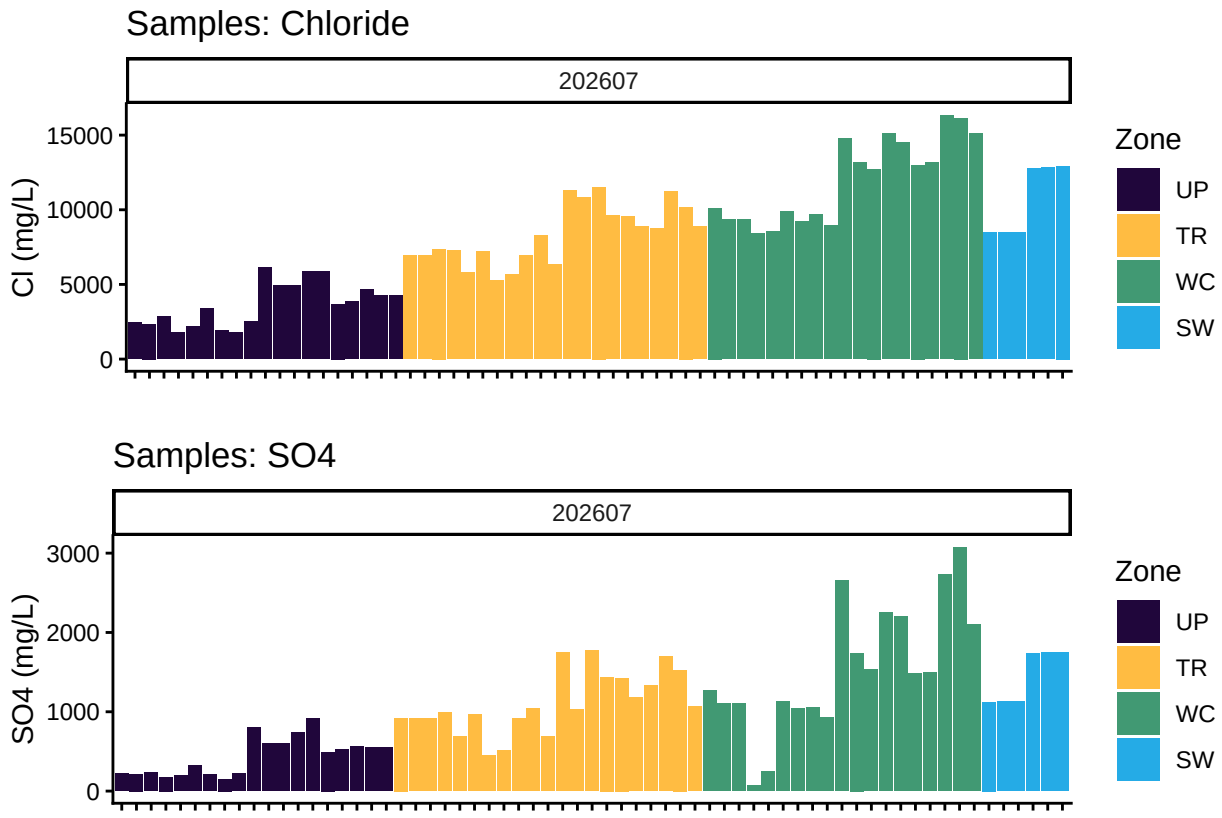
SO4_Conc_flag	Percent_samples
Within_Range	98.461539
adl	1.538461

Table 2: Cl samples

Cl_Conc_flag	Percent_samples
Within_Range	100

1.8 Check to see if samples run match metadata & merge info

```
## All sample IDs are present in metadata.
```


Samples: Chloride
202607
Cl (mg/L)
15000
10000
5000
0
Zone
UP
TR
WC
SW
Samples: SO4
202607
SO4 (mg/L)
3000
2000
1000
0
Zone
UP
TR
WC
SW

#end